

• Referrer Analysis

Problem Statement:

"The store has an organic referral program operating without any formal structure or tracking. The business does not know which age groups are most likely to refer customers, whether referred customers from different age groups convert and stay loyal at different rates, and whether the age of the referrer influences the quality of the referral they give. This analysis investigates referral behaviour across age segments to identify where the referral program is strongest and where it needs attention."

SQL Queries for Referrer Analysis:

```
SELECT
COUNT(DISTINCT c.customer_id)
AS Total_Visited_Customers,
COUNT(DISTINCT CASE WHEN c.referred_by_id IS NOT NULL THEN c.customer_id END)
AS Referral_Customers,
ROUND( COUNT(DISTINCT CASE WHEN c.referred_by_id IS NOT NULL THEN c.customer_id END) * 100 /
COUNT(DISTINCT c.customer_id), 2)
AS Percentage_of_referral_Customers_in_total_visited_customers
FROM customers_details as c;
```

```
WITH unique_referrers AS (
SELECT DISTINCT c.referred_by_id
FROM customers_details as c
),
unique_referrers_referrals AS (
SELECT ur.referred_by_id, c.customer_id as all_customers_id
FROM customers_details as c
INNER JOIN unique_referrers as ur
ON ur.referred_by_id=c.referred_by_id
),
ur_referral_buyers AS (
SELECT urr.referred_by_id, urr.all_customers_id,
i.customer_id as buyer_customer_id,
i.final_bill_amount
FROM unique_referrers_referrals as urr
LEFT JOIN invoices as i
ON urr.all_customers_id=i.customer_id
),
urrb_repeated_referrals AS (
SELECT
urrb.referred_by_id,
urrb.all_customers_id,
urrb.buyer_customer_id,
urrb.final_bill_amount,
rc.customer_id as repeated_customers_id
FROM ur_referral_buyers as urrb
LEFT JOIN repeated_customers as rc
ON rc.customer_id=urrb.all_customers_id
)
```

```
SELECT
COUNT(DISTINCT urrbrr.referred_by_id)
AS Total_Referrers,
COUNT(DISTINCT urrbrr.all_customers_id)
AS Total_Referrals,
COUNT(DISTINCT urrbrr.buyer_customer_id)
AS Buyer_referrals,
ROUND( COUNT(DISTINCT urrbrr.buyer_customer_id)*100/
COUNT(DISTINCT urrbrr.all_customers_id), 2)
AS Referral_Conversion_rate,
SUM(urrbrr.final_bill_amount)
AS Revenue_from_referrals,
ROUND( SUM(urrbrr.final_bill_amount)*100/
(SELECT SUM(final_bill_amount) FROM invoices), 2)
AS Contribution_of_referrals_in_Revenue,
COUNT(DISTINCT urrbrr.repeated_customers_id)
AS Repeated_Referral,
ROUND(COUNT(DISTINCT urrbrr.repeated_customers_id)*100/
COUNT(DISTINCT urrbrr.buyer_customer_id), 2)
AS Referral_Retention_Rate
FROM urrb_repeated_referrals as urrbrr;
```

```
WITH referrers AS (
SELECT
c.referred_by_id
FROM customers_details as c
GROUP BY c.referred_by_id
)
SELECT
CASE
WHEN c.age < 20 THEN "Teenage_Customer"
WHEN c.age BETWEEN 20 AND 30 THEN "20's Customer"
WHEN c.age BETWEEN 31 AND 40 THEN "30's Customer"
WHEN c.age BETWEEN 41 AND 50 THEN "40's Customer"
WHEN c.age BETWEEN 51 AND 60 THEN "50's Customer"
WHEN c.age > 60 THEN "Old_Age Customers (age>60)"
END AS Referrer_Age_group,
COUNT(DISTINCT referrers.referred_by_id ) AS COUNT_Referrals
FROM customers_details as c
INNER JOIN referrers
ON referrers.referred_by_id=c.customer_id
GROUP BY Referrer_Age_group
ORDER BY COUNT_Referrals DESC;
```

```
SELECT
COUNT( DISTINCT c.customer_id ) AS Visited_Customers,
COUNT(DISTINCT i.customer_id) AS Buyer_Customers,
COUNT( DISTINCT CASE WHEN c.referred_by_id IS NOT NULL
AND
i.customer_id IS NOT NULL THEN i.customer_id END)
AS Buyer_referrals,
ROUND( COUNT( DISTINCT CASE WHEN c.referred_by_id IS NOT NULL
AND
i.customer_id IS NOT NULL THEN i.customer_id END) * 100 /
COUNT( DISTINCT i.customer_id ), 2)
AS Percentage_of_referral_customers_in_buyers,
ROUND( (COUNT( DISTINCT CASE WHEN c.referred_by_id IS NOT NULL
AND
i.customer_id IS NOT NULL THEN i.customer_id END))*100/
COUNT( DISTINCT c.customer_id ), 2)
AS Referral_Conversion_Rate
FROM customers_details as c
LEFT JOIN invoices as i
ON c.customer_id=i.customer_id;
```

```
SELECT
'Regular_Customers' AS Type_Customers,
COUNT(DISTINCT c.customer_id) AS Visited_Customers,
COUNT(DISTINCT i.customer_id) AS Buyer_Customers,
ROUND( COUNT(DISTINCT i.customer_id) * 100 /
COUNT(DISTINCT c.customer_id), 2)
AS Conversion_rate_of_non_referral_customers,
COUNT( DISTINCT rc.customer_id) AS Repeated_Customers,
ROUND( COUNT( DISTINCT rc.customer_id) *100 /
COUNT(DISTINCT i.customer_id), 2) AS Retention_rate,
ROUND(AVG(i.final_bill_amount), 2)
AS Average_Revenue_per_invoice
FROM customers_details as c
LEFT JOIN invoices as i
ON c.customer_id=i.customer_id
LEFT JOIN repeated_customers as rc
ON rc.customer_id=c.customer_id
WHERE c.referred_by_id IS NULL
UNION ALL
SELECT
'Referral_Customers',
COUNT(DISTINCT c.customer_id) AS Visited_Customers,
COUNT(DISTINCT i.customer_id) AS Buyer_Customers,
ROUND( COUNT(DISTINCT i.customer_id) * 100 /
COUNT(DISTINCT c.customer_id), 2)
AS Conversion_rate_of_non_referral_customers,
COUNT(DISTINCT rc.customer_id) AS Repeated_referral_customers,
ROUND( COUNT(DISTINCT rc.customer_id)*100/
COUNT(DISTINCT i.customer_id), 2)
AS Retention_rate,
ROUND(AVG(i.final_bill_amount), 2) AS Average_Revenue_per_invoice
FROM customers_details as c
LEFT JOIN invoices as i
ON c.customer_id=i.customer_id
LEFT JOIN repeated_customers as rc
ON rc.customer_id=c.customer_id
WHERE c.referred_by_id IS NOT NULL;
```

```
SELECT
COUNT( DISTINCT c.customer_id)
AS Count_Referral_Customers,
CASE
WHEN c.age < 20 THEN "Teenage_Customer"
WHEN c.age BETWEEN 20 AND 30 THEN "20's Customer"
WHEN c.age BETWEEN 31 AND 40 THEN "30's Customer"
WHEN c.age BETWEEN 41 AND 50 THEN "40's Customer"
WHEN c.age BETWEEN 51 AND 60 THEN "50's Customer"
WHEN c.age > 60 THEN "Old_Age Customers (age>60)"
END AS Age_group
FROM customers_details as c
WHERE c.referred_by_id IS NOT NULL
GROUP BY Age_group WITH ROLLUP;
```

SQL Queries Output of Referrer Analysis

Result Grid		Filter Rows:	Export:	Wrap Cell Content:				
	Total_Referrers	Total_Referrals	Buyer_referrals	Referral_Conversion_rate	Revenue_from_referrals	Contribution_of_referrals_in_Revenue	Repeated_Referral	Referral_Retention_Rate
▶	721	810	381	47.04	4763973.00	10.18	113	29.66

Result Grid				Filter Rows:		Export:		Wrap Cell Content:	
	Total_Visited_Customers	Referral_Customers	Percentage_of_referralL_Customers_in_total_visits						
	8000	810	10.13						

Result Grid		Filter Rows:		Export:	Wrap Cell Content:
	Visited_Customers	Buyer_Customers	Buyer_referrals	Percentage_of_referral_customers_in_buyers	Referral_Conversion_Rate
▶	8000	3721	381	10.24	4.76

Result Grid		Filter Rows:	Export:	Wrap Cell Content:			
	Referrers_Age_group	COUNT_Referrers	Referrals	Buyer_Referrals	Referral_Buyer_Rate	Retain_Buyer_Referrals	Buyer_Referrals_Retention_Rate
▶	20's Customer	125	134	67	50.00	20	29.85
	30's Customer	117	129	55	42.64	19	34.55
	40's Customer	107	115	56	48.70	17	30.36
	50's Customer	97	122	53	43.44	10	18.87
	Old_Age Customers (age>60)	258	288	137	47.57	43	31.39
	Teenage_Customer	17	22	13	59.09	4	30.77
	TOTAL	721	810	381	47.04	113	29.66

Result Grid		Filter Rows:	Export:	Wrap Cell Content:		
	Age_group_Referrals	Count_Referral_Customers	Buyer_Referrals	Referral_Conversion_Rate	repeated_referrals	Referrals_Retention_Rate
▶	20's Customer	130	59	45.38	18	30.51
	30's Customer	134	59	44.03	21	35.59
	40's Customer	133	52	39.10	14	26.92
	50's Customer	108	58	53.70	17	29.31
	Old_Age Customers (age>60)	277	139	50.18	38	27.34
	Teenage_Customer	28	14	50.00	5	35.71
	TOTAL	810	381	47.04	113	29.66

Result Grid

Filter Rows:

Export:

Wrap Cell Content:

	Referrers_Age_group	Teenage_Customers	20s_Customers	30s_Customers	40s_Customers	50s_Customers	Old_Age_Customers
	TOTAL	28	130	134	133	108	277
	Old_Age Customers (age>60)	11	51	43	38	51	94
	50's Customer	5	22	18	28	16	33
	20's Customer	4	17	27	30	16	40
	40's Customer	4	16	21	15	8	51
	30's Customer	3	21	23	19	15	48
	Teenage_Customer	1	3	2	3	2	11

Result Grid									 Filter Rows:		Export:		Wrap Cell Content:	
	Type_Customers	Visited_Customers	Buyer_Customers	Conversion_rate_of_non_referral_customers	Repeated_Customers	Retention_rate	Average_Revenue_per_invoice							
▶	Regular_Customers	7190	3340	46.45	938	28.08	9375.13							
	Referral_Customers	810	381	47.04	113	29.66	9250.43							

Result Grid		Filter Rows:		Export:		Wrap Cell Content:			
	Type_of_Referrers	Buyer_Referrers	Total_Referral_given	Buyer_Referrals	Referrals_Conversion_Rate	Repeated_referrals	Referrals_retention_rate	Revenue_from_referrals	Contribution_in_Revenue
▶	Buyer_Referrers	333	374	168	44.92	49	29.17	2110932.00	4.51
	Non_Buyer_Referrers	388	436	213	48.85	64	30.05	2653041.00	5.67
	Total	721	810	381	47.04	113	29.66	4763973.00	10.18

Referrer Analysis

- About the Referrers

Referral customers represent one of the lowest-cost customer acquisition channels because they require virtually no marketing expenditure while bringing new customers through word-of-mouth. For a local optical store operating without a formal marketing budget or digital advertising presence, referrals represent one of the most natural and cost-free ways where the new customers discover the business.

This section it investigates the entire referral ecosystem from asking who gives referrals, whether those referrals actually buy, how much revenue they generate, and most importantly uncovering a phenomenon that completely challenges the conventional assumption about how word-of-mouth works.

- Scale of Referrals program

Result Grid Filter Rows: Export: Wrap Cell Content:									
Total_Visited_Customers	Total_Referrers	Percentage_of_referrers	Total_Referrals	Buyer_referrals	Referral_Conversion_rate	Revenue_from_referrals	Contribution_of_referrals_in_Revenue	Repeated_Referral	Referral_Retention_Rate
8000	721	9.01	810	381	47.04	4763973.00	10.18	113	29.66

Result Grid Filter Rows: Export: Wrap Cell Content:						
Visited_Customers	Referrers_Customers	Percentage_of_Referrers_Customers	Buyer_Customers	Buyer_referrals	Percentage_of_referral_customers_in_buyers	Referral_Conversion_Rate
8000	721	9.01	3721	381	10.24	4.76

Business Insights:

The analysis reveals that 721 customers (9.01% of all 8,000 visitors) acted as referrers by recommending our store to friends or family. While this represents less than one-tenth of the overall customer base, these referrers generated 810 referral customers (10.13% of the 8000 visitors), demonstrating the significant impact that a relatively small group of satisfied customers can have on business growth.

Out of these 810 referred customers, 381 completed a purchase, resulting in a Referral Conversion Rate of 47.04%. This indicates that nearly one out of every two referred visitors became a paying customer, which is a strong acquisition performance for a completely organic customer source. These 381 referral buyers account for approximately 10.24% of all buyers (3,721) in the business.

Financially, the impact is even more significant. Referral customers generated approximately ₹47.64 lakh in revenue, contributing 10.18% of the store's total revenue. Considering that only 9% of customers became referrers, generating over 10% of total business revenue highlights the high commercial value of referral-based customer acquisition.

However, customer retention remains an area requiring attention. Out of the 381 referral buyers, only 113 became repeat customers, resulting in a Referral Retention Rate of 29.66%. While the initial conversion is strong, retaining these acquired customers presents a clear opportunity for improvement.

Strategic Business Recommendation:

Hence, we must focus more on the referral network functions as a zero-cost customer acquisition channel for the business. Existing customers act as brand advocates by recommending the store to friends and family, increasing awareness without requiring investment in advertising or promotional campaigns.

Rather than investing solely in paid advertising, management should prioritize increasing the number of customers who are willing to recommend the business.

The most sustainable way to achieve this is by consistently delivering an exceptional customer experience throughout the sales journey. Customers are far more likely to refer friends and family when they genuinely trust the business and feel that their interests are prioritized over short-term sales.

If the business can increase the proportion of customers who become active referrers from the current 9% to even 15–20%, referral-driven revenue could increase substantially without proportionally increasing marketing expenditure.

While referral visitors still need to be converted through effective sales and customer service, they represent pre-qualified prospects who arrive with higher initial trust. Strengthening and incentivizing this referral ecosystem has the potential to increase customer acquisition while maintaining a significantly lower Customer Acquisition Cost (CAC) than traditional marketing methods.

Now the question arrives: Who are this Referrers? Whom they Refers (basically referrals)?

Referrers_Age_group	COUNT_Referrers	Referrals	Rate_of_Referrals	Buyer_Referrals	Referral_Buyer_Rate	Retain_Buyer_Referrals	Buyer_Referrals_Retention_Rate	Revenue_Collected
20's Customer	125	134	1.07	67	50.00	20	29.85	778770.00
30's Customer	117	129	1.10	55	42.64	19	34.55	778895.00
40's Customer	107	115	1.07	56	48.70	17	30.36	726073.00
50's Customer	97	122	1.26	53	43.44	10	18.87	508640.00
Old_Age Customers (age>60)	258	288	1.12	137	47.57	43	31.39	1816498.00
Teenage_Customer	17	22	1.29	13	59.09	4	30.77	155097.00
TOTAL	721	810	1.12	381	47.04	113	29.66	4763973.00

Insight 1: Senior Customers are the strongest advocates of the business

Customers aged 60+ emerge as the most valuable advocates within the referral ecosystem.

Compared with every other demographic, this segment contributes:

- Highest number of Referrers
- Highest number of Referral Customers
- Highest number of Retained Referral Customers
- Highest Revenue generated through referrals

This indicates that senior customers are significantly more willing to recommend the store to friends and family, making them the strongest source of organic customer acquisition.

Although their referral-to-buyer conversion rate is slightly below some younger age groups, this is more reflective of the sales conversion process after the referral rather than the referral behaviour itself.

Therefore, the business should continue nurturing this segment through exceptional post-purchase service, relationship management and trust-building initiatives, as they generate substantial long-term value beyond their own purchases.

Insight 2: Middle-aged customer segments demonstrate stable referral behaviour

Customers aged 20–50 display remarkably consistent referral performance across nearly every business metric.

The number of referrals generated, buyer conversion, retained referrals and revenue contribution remain relatively balanced across these age groups, indicating a stable and predictable referral pattern.

This suggests that referral behaviour is not heavily concentrated within a single working-age demographic but is instead broadly distributed. Because these segments represent a large share of the customer base, even small improvements in referral conversion or retention could translate into meaningful revenue growth. Rather than treating these customers differently, the business should apply scalable referral programs and standardized customer experience improvements across all working-age groups.

Insight 3 Teenage customers represent an underpenetrated but high-quality acquisition opportunity

Although teenagers account for the smallest share of referrers and referral customers, they exhibit one of the strongest behavioural profiles once acquired. Compared with other age groups, teenage referrals demonstrate:

- Highest Referral-to-Buyer Conversion Rate
- Lower customer churn
- Strong retention performance

This suggests that the business is not facing a conversion problem within this segment. Instead, the challenge appears to be customer acquisition. In other words,

The business converts teenage customers effectively once they visit the store, but relatively few teenagers enter the acquisition funnel. These highlights an opportunity to expand marketing efforts targeted toward students and young adults through digital marketing, school partnerships, social media campaigns and youth-oriented referral incentives.

Referral Network Analysis by Customer Age Group

Result Grid		Filter Rows:		Export:	Wrap Cell Content:		
	Referrers_Age_group	Teenage_Customers	20s_Customers	30s_Customers	40s_Customers	50s_Customers	Old_Age_Customers
	HULL	28	130	134	133	108	277
	Old_Age Customers (age>60)	11	51	43	38	51	94
	50's Customer	5	22	18	28	16	33
	20's Customer	4	17	27	30	16	40
	40's Customer	4	16	21	15	8	51
	30's Customer	3	21	23	19	15	48
	Teenage_Customer	1	3	2	3	2	11

Result Grid		Filter Rows:	Export:	Wrap Cell Content:			
	Age_group_Referrals	Count_Referral_Customers	Buyer_Referrals	Referral_Conversion_Rate	repeated_referrals	Referrals_Retention_Rate	Revenue_Collected
▶	20's Customer	130	59	45.38	18	30.51	691298.00
	30's Customer	134	59	44.03	21	35.59	732491.00
	40's Customer	133	52	39.10	14	26.92	610299.00
	50's Customer	108	58	53.70	17	29.31	734382.00
	Old_Age Customers (age>60)	277	139	50.18	38	27.34	1839787.00
	Teenage_Customer	28	14	50.00	5	35.71	155716.00
	HULL	810	381	47.04	113	29.66	4763973.00

Key Insight 1: Senior Customers are the Primary Referral Drivers

Customers aged 60+ dominate nearly every referral metric; are as follows:

- Generated the highest number of referrals.
- Referred the highest number of fellow senior customers.
- Produced the highest number of buyer referrals.
- Generated the highest number of retained referral customers.
- Contributed the largest referral revenue (₹18.4 Lakhs).

This reinforces an important business finding observed throughout previous analyses:

Senior customers are not only the store's largest customer segment, but also its strongest brand advocates. Their influence extends beyond their own purchases by continuously bringing new customers into the business.

Key Insight 2: Teenage Referrals Are Small in Volume but High in Quality

Teenage customers represent only a small portion of the referral ecosystem. However, once acquired they demonstrate exceptionally strong performance.

Compared with every other segment, teenage referrals exhibit:

- One of the highest Referral Conversion Rates (~50%)
- The highest Referral Retention Rate (~35.7%)
- Low levels of customer churn

This suggests the business is not facing a conversion problem among teenage referrals. Instead, the challenge lies in customer acquisition. If more teenage customers could be attracted into the referral funnel through school campaigns, digital marketing or social media engagement, this segment could become a meaningful source of long-term growth.

Key Insight 3: Senior Referral Customers Need Better Retention

Although senior customers generate the largest referral revenue, they do not perform equally well in long-term retention. Their referral retention rate remains below several younger segments.

This observation aligns with findings from the earlier Churn Analysis, where older customers also exhibited relatively higher churn behaviour. While the current data cannot determine the exact reason, it indicates that a significant proportion of senior referral customers purchase only once and fail to return.

Recommended actions include:

- Structured follow-up calls after purchase
- Customer satisfaction surveys
- After-sales support
- Reminder campaigns for eye checkups
- Personalized relationship management for senior customers

Improving retention within this high-volume segment is likely to produce a greater business impact than attempting to increase referrals alone.

Insight 4: Senior Customers Receive Referrals Across Nearly Every Age Group

One of the most interesting patterns observed in the referral network is that customers from almost every age segment predominantly refer senior customers (60+) to the store.

This indicates that senior customers are not only referring others themselves, but are also the primary recipients of referrals from family members and acquaintances across multiple age groups.

This pattern suggests that the business has developed a strong reputation and trust in serving senior customers. Families often play an active role in recommending optical care for their parents and grandparents, making senior customers the central segment within the referral ecosystem.

Strategic Business Recommendation:

Context: From an optical retail perspective, this segment naturally requires periodic vision correction due to age-related conditions such as presbyopia and progressive changes in eyesight. As a result, many customers in this age group typically revisit an optical store every 2–3 years for updated prescriptions, replacement lenses or new eyewear. This recurring demand makes customer retention significantly more valuable.

Therefore, the business should prioritize:

- Strengthening customer relationships through personalized service.
- Building long-term trust by recommending products based on customer needs rather than short-term sales incentives.
- Conducting post-purchase follow-ups and collecting customer feedback.
- Scheduling periodic eye-checkup and prescription reminder campaigns to encourage repeat visits.

Protecting and retaining this customer segment should be considered a strategic priority, as today's first-time senior customers are the most likely to become tomorrow's repeat customers, brand advocates and referral generators.

Note: However, the analysis also reveals that a noticeable proportion of senior referral customers do not return after their initial purchase. Since this segment contributes the largest share of referral-driven business, even a small improvement in retention could generate substantial long-term revenue growth.

- Identifying the reasons behind old age customer churn and implementing corrective actions.
- Protect this high-value customer segment.
- Prioritize the retention of this strategic customer segment.
- Safeguarding this segment should be a key business objective.
- Improving retention within this segment offers one of the highest-return opportunities for sustainable revenue growth.

Referral Customers vs Regular Customers

Result Grid		Filter Rows:		Export:		Wrap Cell Content:	
Type_Customers	Visited_Customers	Buyer_Customers	Conversion_rate_of_non_referral_customers	Repeated_Customers	Retention_rate	Average_Revenue_per_invoice	
Regular_Customers	7190	3340	46.45	938	28.08	9375.13	
Referral_Customers	810	381	47.04	113	29.66	9250.43	

Business Insight:

At first glance, referral customers appear to perform slightly better than regular customers across key business metrics. However, the difference is relatively small.

Now the Question Arrives what's next? What decisions we should make further?

"Based on the analyses performed in this project, two strategic opportunities appear to offer the greatest potential for future business growth:

1. Expanding Customer acquisition among younger demographics

The analysis consistently shows that teenage customers exhibit:

- One of the highest conversion rates
- Strong retention performance
- Lower customer churn

The current limitation is not conversion but customer volume. Very few teenage customers enter the business, despite performing exceptionally well once acquired.

2. Improving the retention of senior customers (Age above 60)

The analysis also identifies senior customers as the most strategically valuable customer segment.

They contribute the highest:

- Revenue
- Repeat customers
- Referral generation
- Referral revenue
- Long-term business value

Key Insight – Non-Buyer Referrers Are an Unexpected Growth Driver

Result Grid Filter Rows: Export: Wrap Cell Content:										
Type_of_Referrers	Buyer_Referrers	Total_Referral_given	Referrals_Rate	Buyer_Referrals	Referrals_Conversion_Rate	Repeated_referrals	Referrals_retention_rate	Revenue_from_referrals	Contribution_in_Revenue	
Buyer_Referrers	333	374	1.12	168	44.92	49	29.17	2110932.00	4.51	
Non_Buyer_Referrers	388	436	1.12	213	48.85	64	30.05	2653041.00	5.67	
Total	721	810	1.12	381	47.04	113	29.66	4763973.00	10.18	

One of the most surprising findings from this analysis is that customers who have never purchased from the store are recommending the business almost as actively as existing buyers.

Although the business has a larger number of Non-Buyer Referrers (388) than Buyer Referrers (333), both groups generate referrals at an almost identical rate of 1.12 referrals per referrer. More importantly, the quality of these referrals remains remarkably similar throughout the customer journey.

Referrals generated by non-buyers achieved a 48.85% conversion rate, compared to 44.92% for buyer referrers, while referral retention rates remained almost identical (30.05% vs. 29.17%). Consequently, both groups contributed nearly the same business value, with buyer referrers generating approximately ₹21.1 lakh in revenue and non-buyer referrers contributing approximately ₹26.5 lakh.

Collectively, the referral ecosystem generated ₹47.64 lakh, accounting for 10.18% of total business revenue without any direct marketing expenditure.

The most important takeaway from this analysis is not the revenue itself, but the question it raises:

“Why are customers who have never purchased from us still recommending our business?”

This behaviour suggests that purchasing is not the only factor influencing word-of-mouth referrals. Several business HYPOTHESIS / Explanations are possible:

- The customer may have had a positive consultation or store experience despite not making a purchase.
- Price-sensitive visitors may have chosen not to buy immediately but still trusted the store enough to recommend it to friends or family.
- Existing reputation, service quality, staff behaviour, or brand trust may be creating positive impressions independent of sales.

This represents an opportunity for further investigation through customer feedback or surveys.

Understanding why non-buyers become advocates could help the business strengthen referral acquisition while also improving conversion among these visitors.

Referrer Analysis Section Summary

The Referrer Analysis demonstrates that the store's referral ecosystem is a significant contributor to long-term business growth. Although only 721 customers (approximately 9% of the total customer base) acted as referrers, they generated 810 referral customers, of which 381 became buyers, contributing ₹47.64 lakh, or approximately 10.18% of the store's total revenue. This highlights referrals as one of the most valuable acquisition channels, achieved without direct marketing expenditure.

The analysis also revealed that buyer and non-buyer referrers exhibit nearly identical referral behaviour, generating approximately 1.12 referrals per referrer on average. Surprisingly, non-buyer referrers performed almost as well as buyer referrers in terms of referral generation, conversion, retention, and revenue contribution. This suggests that customer advocacy is driven not only by purchases but also by the overall customer experience and perception of the business.

Demographic analysis further showed that customers above 60 years of age form the backbone of the referral network. They contribute the highest number of referrers, referrals, repeat referral customers, and referral revenue. Given that this customer segment naturally revisits optical stores every 2–3 years due to age-related prescription changes, retaining these customers should be treated as a strategic priority. At the same time, teenage customers, despite their relatively small volume, demonstrated the highest referral conversion and retention rates, indicating a promising opportunity for future customer acquisition through digital marketing and targeted engagement.

Overall, the findings suggest that the referral program is already performing effectively and represents a highly cost-efficient growth channel. Future business efforts should focus on strengthening customer experience, increasing referral participation, improving referral customer retention, and expanding acquisition among younger customer segments while continuing to nurture the highly valuable senior customer base.

Final Business Recommendations

- **Strengthen customer experience** by delivering accurate prescriptions, transparent consultations, and exceptional after-sales support, as satisfied customers become the business's most effective marketers.
- **Retain senior customers** through periodic follow-ups, reminder campaigns, and proactive eye-checkup programs, recognizing their high lifetime value and recurring vision care needs.
- **Expand referral participation** by encouraging satisfied customers to recommend friends and family through structured referral initiatives.
- **Target younger demographics** using social media and digital marketing, leveraging their strong conversion and retention characteristics to build the next generation of loyal customers.
- **Investigate churn among senior referral customers** by collecting feedback from one-time buyers and implementing corrective actions to improve long-term retention.